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CABLE LOCK-OFF BLOCK WITH ADJUSTABLE FLAT WEDGES FOR REPAIRING A PLURALITY OF POST-TENSIONED TENDONS

Abstract

A post-tensioning system having a cable lock off block using adjustable wedges for repairing a plurality of post-tensioned tendons. The lock off block uses a wedge body shape constructed and arranged to eliminate inherent interference created when using traditional conical wedges and conical wedge cavities. To allow for wedge adjustability, the lock off block body comprises an elongated single tapered slot. Use of the current wedge construction, in combination with a block off body having elongated tapered slot, allows a user to position individual wedges directly over each individual tendon, even if the cables are touching each other.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates to post-tensioning systems; more particularly, to a post-tensioning system having a cable lock-off block using adjustable wedges for repairing a plurality of post tensioned tendons.

BACKGROUND OF THE INVENTION

[0002] Concrete, a commonly used building structure material, can carry substantial compressive loads, but is unable to carry significant tensile loads. To overcome this, adding reinforcing steel bars, wires or cables to concrete is required to increase its tensile carrying ability in a concrete structure. Basic reinforcement methods in concrete structures include: conventionally reinforced structures and pre-stressed concrete structures.

[0003] As described in U.S. Pat. No. 9,315,998 ('998 Patent), in conventionally reinforced concrete structures, deformed steel bars, called reinforcing bars, are placed in tensional stress areas of the concrete members, thus allowing the steel to carry the tensile forces and the concrete to carry compressive forces.

[0004] In pre-stressed concrete structures, the method of pre-stressing can be defined as the application of a pre-determined force or moment to a structural member, in such manner that the combined internal stresses in the member, resulting from this force or moment and from any anticipated condition of external loading, is being confined within specific limits. Pre-stressed concrete is the result of applying this principle to concrete structural members, and eliminating or materially reducing the tensile stresses in the concrete. In this type of structure, the type of reinforcement methods separates the type of structures into two basic groups: pre-tensioned structures, and post-tensioned structures.

[0005] In pre-tensioned structures, reinforcing rods, cables or strands of high tensile strength wires (called tendons) are first pre-stretched to a certain pre-determined amount, and then high-strength concrete is placed around the reinforcing rods to form the concrete member. Once the concrete has set, pre-tensioned rods are released to introduce a compressible force into the member while the concrete holds the steel reinforcement in a tight bond, preventing slippage and sagging.

[0006] The tendons used in pre-tensioned construction must be relatively small in diameter because the bond stress between the concrete and the tendon is relied upon to transfer the stress from the tendon to the concrete. Pre-tensioning is mostly performed within individual concrete members at their manufacturing plant.

[0007] In post-tensioned structures, reinforcing rods, cables or strands of high strength wires (called tendons) are draped loosely to a profile as determined by structural analysis, and high-strength concrete is placed around them. The tendons are usually encased in a flexible plastic protective hose (called sheath or duct) to prevent the tendon from bonding to the concrete during placement and curing of the concrete. The protective sheathing remains in the structure. After the concrete has reached its pre-determined strength, the tendons are then stretched by hydraulic jacks and are securely anchored into place by some type of device, mostly at their end locations. These devices are referred to as end anchorages. The end anchorages, together with the special jacking and grouting equipment used in accomplishing the post-tensioning by one of the several methods, are referred to as post-tensioning systems.

[0008] In some cases, the void between the tendon and the sheath is filled with grout. In this manner, the tendon becomes bonded to the concrete section and corrosion of the steel tendon is prevented. In some other cases, the tendon is coated with grease prior to placement into a protective sheathing. Tendons of this type are not pressure grouted after stressing. This type of post-tensioning is usually referred to as an un-bonded post tensioning system. Post-tensioning of tendons is generally performed at the construction site.

[0009] In some instances, the metal components of the post-tensioning systems may become exposed to external sources of chlorides in service, such as de-icing salts, brackish water, seawater,

or spray from these sources. There have been corrosion problems with either type of pre-and post-tensioning systems. However, certain aspects of corrosion of un-bonded single strand tendons are unique, and the end effects of corrosion of un-bonded single strand tendons are, in several respects, different from those of bonded conventional reinforcing or other post-tensioning systems. Thus, the methods for evaluating and repairing corrosion of single strand tendons are also different in some respects. For example, since the tendons are largely isolated from the surrounding concrete, they may not be affected by deleterious materials, such as chlorides and moisture in the concrete. However, they also are not completely protected by the surrounding concrete, and can corrode if water gains access to the inside of the sheathing or anchorage and the grease protection is inadequate. Measures taken to repair and protect the surrounding concrete may not repair or reduce deterioration of the pre-stressing steel where corrosion has been initiated. The tendons usually require separate evaluation and repair.

[0010] The use of un-bonded tendons became more common during the late 1950s and early 1960s as design standards and materials standards were established. Due to their advantage over conventional type structures (shorter construction time, additional living space gain by reducing slab thickness, and savings in overall cost), the use of post-tensioning gained more popularity during the late 1960s and 1970s and became one of the common type structural systems for many applications. In addition to their use in various building structures, un-bonded post-tensioning systems were used in parking structures, slab-on-grade, and nuclear power structures. The earliest incidents of corrosion of un-bonded tendons began to surface during the 1970s, since the grease and the sheathing materials (leading to cable corrosion) used at that time period did not provide proper protection for corrosion. In the early 1980s, the Post-Tensioning Institute (PTI) recognized the structural implications of corrosion and began to implement measures to increase the durability of un-bonded post-tensioning systems. In 1985, PTI published the first performance standards for un-bonded tendons and included standards for grease (corrosion-inhibiting hydrophobic grease). The 1989 edition of American Concrete Institute (ACI 318), "Building Code Requirements for Reinforced Concrete", made changes to include protection measures for the tendons and the quality of the concrete from the environmental conditions that would promote corrosion. Structures built prior to the adaptation of these new standards are presently experiencing corrosion problems and are in need of repair. Perhaps more importantly, mild steel rebar located behind anchors are susceptible to corrosion. As a major cause of damage, particularly for elevated commercial and residential structures, these corroded mild steel rebar warrants splice repairs.

[0011] When a tendon has been inadvertently cut and has no tension, the tendon, along with its end anchorages, is usually replaced. The original anchors may be reused, but dislodging the old wedges is sometimes difficult and the anchors can be damaged in the process. It is usually advisable to replace the anchors with new ones since this provides the opportunity to improve the system's durability. Once free of its anchorage, strand extraction is normally not difficult. In some cases, a jack can be used to pull the strand out. This method, however, while reliable, is slow. Usually the loose tendon is pulled out by hand or with the assistance of a come-along or a vehicle. When a tendon is damaged, or when corrosion damage is known or believed to be localized, repairs are often made by replacement of a part of the tendon, either between anchorages or on one side of the damage. If the un-damaged portions of the tendons are in good condition, the damaged section of tendon is cut away and a new piece of tendon is spliced onto the ends of the original tendon using splice couplers. The old anchors may be reused as long as the tensional force in the tendon is maintained, and if the existing wedges are not unlocked. If the tendon has been de-tensioned, then the wedges are always replaced.

[0012] In most cases, the tendon damage is localized and can be determined by investigation prior to repairs. Usually un-bonded tendons are damaged close to their end anchorages and the remaining portions of the tendons are still in good condition. In these cases, the tendons are temporarily anchored (locked-off) by installing temporary anchors at locations where the tendons are still in

good condition; the damaged end of the tendons are then cut and removed. A new tendon is spliced to the end of each existing tendon and the existing anchor is replaced with a new one. In order to lock the tendons with their internal tensional force intact, the existing tendon is usually exposed through a small chipped hole (chipped in concrete), its sheathing is removed, and a temporary-anchor with two sets of wedges is installed around the tendon. The tensional force at the damaged side of the tendon is then gradually released by chipping the concrete behind the existing end-anchor and, by doing so, transferring the tensional force onto the new temporary-anchor. The entire operation herein is referred as "tendon lock-off". The tendon lock-off operation is usually simple and unproblematic where the existing tendons are separate from each other, at locations where they are near to their original anchors or located in a uniform fashion. In post-tension band-lines (where post tensioning tendons are bundled to go over column lines and provide support for transfer-directional uniformly placed tendons), it is common practice to bundle several tendons together for ease of construction. Usually, four tendons per flat-bundle is the maximum recommended for floor slab construction. This limitation is pursued for two reasons. First, apart from poor consolidation, there is an increased potential of delamination at high and low profile points, and second, there is an increased probability of blow-outs at locations of horizontal curvatures due to outer strands riding over the inner ones. Unfortunately, in common practice, this recommendation is not always followed and more than four tendons are bundled together in a flat fashion. The repair of these banded tendons becomes difficult due to insufficient space between the tendons. The repair operation requires a larger exposure area for the placement of temporary-anchors, and/or most of the time, and instead of de-tensioning and repairing a single damaged tendon, it requires replacement of entire banded tendons.

[0013] The 1998 patent provided a device that temporarily transfers the existing tension load to a new location for the purpose of repairing the cables, particularly multiple cable bundles. While an effective device for repairing a plurality of post tensioned tendons, the device described in the 1998 patent had several shortcomings, including the wedges being prevented from securing individual cables that are immediately adjacent to each other.

SUMMARY OF THE INVENTION

[0014] A post-tensioning system having a cable lock-off block using adjustable (and adjustably positionable, including being placed in a position and/or being movable in a linear direction, left or right) wedges for repairing a plurality of post-tensioned tendons. The present invention improves upon existing post-tensioned hardware which uses circular, cone shaped sets of wedges to secure the cable to the given piece of hardware. For "½" cables, the prior art teaches using two-piece or three-piece wedge sets, that are typically 1" +/- diameter.

[0015] In repairing tension steel cables associated with concrete structures, the tension load must be temporarily removed or relocated beyond the repair area. Following the repair, reestablishment of the cable tension load back to its original span is required. The post-tensioning system in accordance with embodiments of the present invention includes a cable lock-off block using adjustable wedges for repairing the plurality of post-tensioned tendons. The cable lock-off block temporarily transfers the existing tension load to a new location for repairing the cables. For concrete structures, steel cable bundles are typically spaced so that the adjacent cables are almost touching each other. For concrete structures using the standard cable wire, i.e. a 7-wire cable, 1" in diameter, the center to center spacing between each cable can be as little as 0.70" +/-.

[0016] Traditional lock-off blocks use 1" diameter round, cone shaped wedges to secure each cable. Given the potential minimum center to center cable spacing of 0.70" +/- between individual cables in a bundled condition, the prior art 1" diameter wedges prevent securing individual cables that are immediately adjacent to each other. This physical limitation means that the minimum spacing between wedge cavities in the device cannot be less than 1.40" center to center. To overcome this shortcoming, the prior art device addressed this by spacing wedge cavities far enough apart to allow every other cable to be secured and providing enough space in between for the next

immediately adjacent cable. However, this configuration requires stacking of lock-off blocks on top of each other for securing the skipped cables.

[0017] Embodiments of the present device avoid these shortcomings by using an improved lock-off block in temporarily transferring an existing tension load to a new location in repairing the cables associated with concrete structures. The improved lock-off block uses a new wedge body having a flat wedge design, constructed and arranged to eliminate the inherent interference created when using traditional conical wedges and conical wedge cavities. To allow for wedge adjustability, the lock-off block body comprises an elongated single tapered slot. Use of the flat wedge design, in combination with a lock-off block body having an elongated tapered slot, allows a user to position individual wedges directly over each individual cable, even if the cables are touching each other.

[0018] Embodiments of invention, therefore, the present provide a user a system which secures bundled cables using a single device, even when the center-to-center cable spacing is $0.70'' \pm$. As such, in comparison to the required minimum wedge cavity spacing of the existing prior art block off devices of $1.40''$, center to center, the present system is constructed and arranged to permit a minimum center to center cable spacing of $0.70'' \pm$. Such differences, while small, result in removing obstacles that exist in the field when attempting to retroactively relocate the cable forces for a repair.

[0019] Accordingly, it is an objective of the invention to provide post-tensioning systems for repairing a plurality of post tensioned cables/tendons.

[0020] It is a further objective of the invention to provide a post-tensioning system having a cable lock-off block using adjustable wedges for repairing a plurality of post-tensioned tendons.

[0021] It is yet another objective of the invention to provide a post-tensioning system having a cable lock-off block using unique, flat wedges for repairing a plurality of post-tensioned tendons.

[0022] It is a still further objective of the invention to provide a post-tensioning system for repairing a plurality of post-tensioned tendons having a cable lock-off block using wedges constructed and arranged to eliminate the inherent interference created when using the post-tensioning system utilizing conical wedges.

[0023] It is a further objective of the invention to provide a post-tensioning system for repairing a plurality of post-tensioned tendons having a cable lock-off block having an elongated single tapered slot.

[0024] It is yet another objective of the invention to provide a post-tensioning system for repairing a plurality of post-tensioned tendons having a cable lock-off block having wedges constructed and arranged to eliminate the inherent interference created when using the post-tensioning system utilizing conical wedges, in combination with an elongated single tapered slot.

[0025] It is a still further objective of the invention to provide a post-tensioning system for repairing a plurality of post-tensioned tendons having a cable lock-off block having wedges constructed and arranged to permit individual wedges to be located directly over each individual cable, even if the cables are touching each other.

[0026] It is a further objective of the invention to provide a post-tensioning system for repairing a plurality of post-tensioned tendons having a cable lock-off block having wedges constructed and arranged to allow the user to secure bundled cables using a single device, even when the center-to-center cable spacing is $0.70'' \pm$.

[0027] Other objectives and advantages of this invention will become apparent from the following description taken in conjunction with any accompanying drawings wherein are set forth, by way of illustration and example, certain embodiments of this invention. Any drawings contained herein constitute a part of this specification, include exemplary embodiments of the present invention, and illustrate various objects and features thereof.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0028] FIG. 1 is a perspective view of an illustrative embodiment of a cable lock-off block using adjustable wedges for repairing a plurality of post-tensioned tendons, shown engaged with three tendons;

[0029] FIG. 2 illustrates the positioning of a pair of adjustable wedges engaging with a post-tensioned tendon;

[0030] FIG. 3 is a top view of the cable lock-off block, shown engaged with three post-tensioned tendons and including a block insert;

[0031] FIG. 4 is a bottom view of the cable lock-off block illustrated in FIG. 3;

[0032] FIG. 5 is a top view of the cable lock-off block, shown engaged with three post-tensioned tendons;

[0033] FIG. 6 is a bottom view of the cable lock-off block illustrated in FIG. 5;

[0034] FIG. 7 is a front perspective view of the cable lock-off block, shown without post-tensioned tendons;

[0035] FIG. 8 is a back-perspective view of the cable lock-off block illustrated in FIG. 7;

[0036] FIG. 9 is a left side view of the cable lock-off block illustrated in FIG. 7;

[0037] FIG. 10 is a right-side view of the cable lock-off block illustrated in FIG. 7;

[0038] FIG. 11 is a bottom perspective view of the cable lock-off block illustrated in FIG. 7;

[0039] FIG. 12A is a perspective view of an illustrative embodiment of a wedge body;

[0040] FIG. 12B is an alternative perspective view of the wedge body shown in FIG. 12A;

[0041] FIG. 13 illustrates a pair of wedges, shown without a post-tensioned tendon;

[0042] FIG. 14 is a side view of the wedge body shown in FIG. 12A;

[0043] FIG. 15 illustrates an embodiment of a pair of wedges, shown without a post-tensioned tendon;

[0044] FIG. 16 illustrates three pairs of wedges, shown without post-tensioned tendons;

[0045] FIG. 17 illustrates three pairs of prior art rounded wedges, shown without post-tensioned tendons;

[0046] FIG. 18 is a cross-sectional view taken along lines A-A of FIG. 3;

[0047] FIG. 19 is a cross-sectional view taken along lines B-B of FIG. 3;

[0048] FIG. 20 is a cross-sectional view showing one half of the wedge pairs with tendons;

[0049] FIG. 21 is a cross-sectional view, illustrating full/both wedges of the wedge pair engaged with tendons;

[0050] FIG. 22 is a perspective view of an illustrative example of a insert/wedge plate;

[0051] FIG. 23 illustrates two cable lock-off blocks arranged one in front of another, thereby providing for multiple tendons; and

[0052] FIG. 24 is a cross sectional representation of two cable lock-off blocks arranged one in front of another.

DETAILED DESCRIPTION OF THE INVENTION

[0053] While the present invention is susceptible of embodiment in various forms, there is shown in the drawings and will hereinafter be described a presently preferred, albeit not limiting, embodiment with the understanding that the present disclosure is to be considered an exemplification of the present invention and is not intended to limit the invention to the specific embodiments illustrated.

[0054] Referring to FIGS. 1-12, an illustrative example of a cable anchor block for repairing a plurality of post-tensioned tendons using adjustable (and adjustably positionable) wedges, referred to generally as an adjustable lock off 10, is provided. In use, the adjustable lock off 10 temporarily transfers an existing tension load to a new location for repairing cables within a concrete structure.

The adjustable lock off **10** can retain and anchor multiple post-tensioning cables (banded tendons that are laid side by side in a flat bundled fashion and are under full tension) during their repair operations. The adjustable lock off **10** is configured to be used with un-bonded post-tensioned cable reinforcement, such as ½" un-bonded post-tensioned cable reinforcement, or 7-wire cable, "½" in diameter, and tensioned to an individual cable load of 27,000 pounds each. The adjustable lock off **10** is intended for interaction with multiple cable/cable bundles.

[0055] Referring specifically to FIG. **1**, the adjustable lock off **10** is shown with four, multiple (seven wire) wire steel cables (or reinforced rods, or strands of high tensile wires, referred to generally as a tendon **12**, and individually as tendon **12A**, tendon **12B**, tendon **12C**, and tendon **12D**). The tendons **12** are shown throughout without a protective sheath, which is usually removed prior to placement into the lock off **10**. The adjustable lock off **10** comprises an anchor body **14**, a slotted portion **16**, and one or more adjustable post-tensioned cable engaging devices **18** constructed and arranged to secure the individual tendons **12** to the adjustable lock off **10**. The one or more adjustable post-tensioned cable support engaging device **18** allows a user to move and position the one or more adjustable wedges (described later) anywhere within the slotted portion **16**. This adjustability allows for engagement and interaction of multiple tendons **12** without having to reposition or move the entire adjustable lock off **10**. As illustrated in FIG. **3** and FIG. **4**, the adjustable lock off **10** is shown with the one or more adjustable post-tensioned cable support engaging device **18** engaged or interacting with tendon **12A**, tendon **12B**, and tendon **12D**. Tendon **12C** is not engaged or interacting with one or more adjustable post-tensioned tendon engaging device **18** (tendon **12C** is not locked off).

[0056] Referring to FIG. **5** (top view) and FIG. **6** (bottom view), the adjustable lock off **10** is illustrated interacting or engaging with multiple tendons **12**. Specifically, the adjustable lock off **10** is shown with an adjustable post-tensioned tendon engaging device **18** engaged or interacting with tendon **12A**, an adjustable post-tensioned tendon engaging device **18** engaged or interacting with tendon **12B**, and an adjustable post-tensioned tendon engaging device **18** engaged or interacting with tendon **12C**. None of the one or more adjustable post-tensioned tendon engaging devices **18** engage or interact with tendon **12D** (cable **12D** is not locked off).

[0057] Referring to FIGS. **7-11**, the adjustable lock off **10**, shown without the one or more adjustable post-tensioned tendon engaging devices **18**, comprises the anchor body **14** having a first surface or wall **22** and a second, opposing surface or wall **24**. The anchor body may be operatively arranged to accommodate one half of a total accumulated tension force or shear force applied by the plurality of post-tensioned tendons **12**. The second opposing surface or wall **24** is positioned against a concrete surface and is not used for temporary anchoring of tendons. The slot or channel **16**, preferably running through the center of the anchor body **14**, starts at a closed end **21** of the anchor body **14** and extends through an outer edge **23**, forming an open end **25**.

[0058] The extended slot or channel **16** provides an opening for which the multiple tendons **12** can be inserted and/or rest within. Running through the center of the anchor body **14** is an optional center recess **26** (with the slot or channel **16** running centrally therethrough) and opening **28** which exposes at least a portion of an interior **30**. The optional center recess **26** allows for multiple adjustable lock off units **10** to be stacked (one on top of another, see FIG. **23** and FIG. **24**) without interference of the adjustable post-tensioned tendon engaging device **18** (wedges **56**) protruding out. The recess **26** may be centered transversally about a center of the anchor body **14** and offset longitudinally to accommodate a bolt hole (**40**, **42**) and a bolt **44** placed perpendicular to the slot or channel **16**.

[0059] The slot or channel **16**, along with the opening **28** shapes the anchor body **14**, which in this illustrative example forms a generally U-shaped form. Accordingly, the anchor body **14** may include a base portion **32**, a first elongated section or arm **34**, and a second elongated section or arm **36**; the first elongated section or arm **34** extending in a direction away from the base portion **32**. The second elongated section or arm **36** extends in a direction away from the base portion **32**.

Both the first elongated section or arm **34** and the second elongated section or arm **36** have a narrower surface profile or height, see arrow **35**, FIG. 7, compared to the surface profile or height of the first surface or wall **22** or the second, opposing surface or wall **24**, see arrow **37**, FIG. 7. [0060] The first elongated section or arm **34** and the second elongated section or arm **36** are preferably arranged in a generally parallel orientation relative to each other and are separated by the slot or channel **16**. The separation forms a gap or **38** opposite the base portion **32**. Both the first elongated section or arm **34** and the second elongated section or arm **36** each contain a bolt opening, **40**, **42** (See FIG. 7). The bolt openings **40**, **42** are sized and shaped to receive and hold a fastening device, such as a bolt **44** (shown with a threaded body **45**) with a nut **46** and a washer **48** (see FIG. 3 and FIG. 18). The bolt **44** may be centered about mid depth of the main anchor **14** and is placed perpendicular to a removable object/insert, illustrated herein as an insert plate/shim **49** (about the center of the insert plate), see FIG. 18 (FIG. 3, lock off unit **10** shown with insert; FIG. 22, unassembled view). The bolt **44** is operatively arranged to accommodate a total accumulated shear force applied by the plurality of post-tensioned tendons **12**.

[0061] The insert plate **49** is sized and shaped to fit within the area of slot or channel **16** which separates portions of the first elongated section or arm **34** and the second elongated section or arm **36**. The insert plate **49** embodiment shown may include a body **51** having a center opening **53** for receiving the bolt **44**. The body may have a first wall or surface, an opposing second wall or surface **55**, a first side wall or surface **57**, and a second side wall or surface **59**. The second side wall or surface **59** may include a cut-out portion **61**, defining a recessed ledge **63**, thus forming seat **65**. The second side wall or surface **59** may include a curved surface **67**. Alternately, the second side wall or surface **59** may not include the curved surface **67** or the cut-out portion **61**. Placed within, the insert plate **49** blocks the end of the slot or channel **16**, thus retaining the multiple post-tensioning tendons **12** in place during re-tensioning operations. The insert plate **49** is secured to the adjustable lock off **10** via the bolt **44**. Accordingly, the insert plate **49** may be configured to match the exact shape of the anchor body **14** cross-section.

[0062] The center recess **26** and opening **28** are constructed and arranged to hold and house one or more of the adjustable post-tensioned cable engagement devices **18**. The center recess **26** and opening **28** acting to enlarge at least a section of the slot or channel **16**. The interior of the center recess **26** may be defined by a ledge or upper wall **50** positioned below the opening **28**. The ledge or upper wall **50** defines a seat for which at least a portion of the one or more adjustable post-tensioned cable engagement devices **18** rest thereon, or move or slide thereupon. A second or side wall **52** frames or is continuous around the ledge or upper wall **50**, and extends in a direction upwardly from the ledge or upper wall **50**. An inner wall **54** is preferably angled inwardly (toward the center), thus defining an inwardly tapered wall or interior tapered section. Preferably, the degree of tapering is defined by or mirrors or corresponds to the degree of tapering associated with the one or more adjustable post-tensioned cable engagement devices **18** (mirroring the angle of the flat wedges as described later).

[0063] Referring to FIGS. 12A-14, the adjustable post-tensioned cable engagement device **18** is shown as a one (or a pair of) wedge body **56**. The wedge body **56** comprises a first or top wall or surface **58**, an opposing second wall or surface **60**, a first side wall or surface **62**, and a second side wall or surface **64**. The wedge body **56** is further defined by an outer wall or surface **66** (relative to contacting or engaging with a portion of the inner wall **54**) and an inner wall or surface **68** (relative to contacting or engaging with a tendon **12**). The inner wall or surface **68** includes a curved inner surface **70** (illustrated as a half circle shape), with the curvature mirroring or corresponding to the curvature or diameter of the tendon **12**. The first or top wall or surface **58** may also include a curved surface **72**, with the curvature mirroring or corresponding to the curvature or diameter of the tendon **12**. The inner wall or surface curved inner surface **70** and the first or top wall or surface curved surface **72** form a first portion of a cable enclosure. The second portion of a cable enclosure is formed by a second wedge body **56**, thus forming a completed adjustable post-tensioned tendon

engagement device **18**.

[0064] The curved inner surface **70** terminates in an inner wall or surface first side edge **74** and an inner wall or surface second side edge **76**. The inner wall or surface first side edge **74** comprises a surface **78** that runs the entire length of the wedge body **56** (from the first wall or surface **58** to the opposing second wall or surface **60**) and is generally flat or linear, i.e. no slanting, formed of an angled surface, or tapered surface with a taper angle. The inner wall or surface second side edge **76** comprises a surface **80** that runs the entire length of the wedge body **56** (from the first wall or surface **58** to the opposing second wall or surface **60**) and is generally flat or linear, i.e. no slanting, formed of an angled surface, or tapering.

[0065] Accordingly, the inner wall or surface **68** forms or defines a flat or non-tapered side of the wedge body **56**. As shown in FIG. **12B**, the wall or surface **66** is defined by an angled wall or surface, this providing a downwardly (towards the opposing second wall or surface **60**) tapering. Accordingly, the outer wall or surface **66** forms or defines a tapered side of the wedge body **56**. The degree of tapering of outer wall or surface **66** corresponds to the degree of tapering associated with the inner wall **54**, thus securing an engaged (and locked) tendon **12**, see FIG. **1**. FIG. **13** illustrates a pair of engaged wedges, **56-1** and **56-2**, shown without a post-tensioned tendon.

[0066] In use, the ability to retain and anchor a post-tensioning cable **12** is accomplished using two wedge bodies **56**. Since the slot or channel **16**, recessed portion **26** and opening **28** of the adjustable lock off **10** forms open structures, i.e. something can be inserted within, and do not including pre-existing openings or wedge receiving areas, each wedge body **56** is freely positioned or moved anywhere within the slot or channel **16**, the recessed portion **26**, and the opening **28**. As shown in FIG. **2** (and FIG. **19**., cross sectional view of the wedge bodies **56A** and **56B** locked to a tendon **12** within at least a portion of the slot or channel **16**, recessed portion **26** or opening **28**), a first wedge body, referred to as wedge body **56A**, is positioned to fit within a portion of the slot or channel **16**, recessed portion **26** or opening **28** associated with the first elongated section or arm **34**. A second wedge body, referred to as wedge body **56B**, is positioned to fit within a portion of the slot or channel **16**, recessed portion **26**, or opening **28** associated with the second elongated section or arm **36**. FIG. **20** (cross-sectional view, showing half/one wedge of wedge pairs)) and FIG. **21** (cross-sectional view, showing full/both wedges of the wedge pair further illustrate the positioning of the adjustable lock off **10**, shown with a concrete surface **100**, with the wedges **56** locked or unlocked to tendons **12**, and positioned within at least a portion of the slot or channel **16**, recessed portion **26** or opening **28**.

[0067] Referring to FIGS. **15-16**, embodiments of illustrative examples of the two wedge bodies, referred to as wedge body **56A** and wedge body **56B**, are shown aligned together, but with the post-tensioning tendon **12** removed. In this alignment, the distance between the top wall or surface **58** and the opposing bottom wall or surface **60** is a distance of 1.53 inches. The distance between the edge of the top wall or surface **58** of wedge body **56A** to the corresponding edge of the top wall or surface **58** of wedge body **56B** is 1.666 inches. The distance between the edge of the bottom wall or surface **60** of wedge body **56A** to the corresponding edge of the bottom wall or surface **60** of wedge body **56B** is 0.9200 inches. The distance between the center line **80** of the wedge body **56A** and the center line **82** of the wedge body **56B** is 0.50 inches. The degree of tapering of the outer wall or surface **66** of the wedge body **56A** and the outer wall or surface **66** of the wedge body **56B** is 13.712 degrees. While the embodiments shown in FIGS. **15-16** define various measurements and sizes, such measurements and sizes are illustrative only and not meant to be limiting.

[0068] The use of the two wedge bodies having a tapered side and a flat or linear side in accordance with the present invention provides a unique benefit over existing post-tensioning systems. FIG. **16** illustrates three adjustable post-tensioned cable engagement devices: **18A** (having wedge bodies **56A** and **56B**), **18B** (having wedge bodies **56A** and **56B**), and **18C** (having wedge bodies **56A** and **56B**), each designed to secure multiple tendons **12** with minimum spacing. Here, the distance between the three adjustable post-tensioned cable engagement devices **18A**, **18B**, and

18C being ± 2.0 inches.

[0069] FIG. 17 illustrates a prior art post-tensioned cable support system using a multi-cable lock-off anchor, see U.S. Pat. No. 9,315,998. As described in the 1998 Patent, the tensioned cable engaging devices **19A**, **19B** and **19C** utilize 1" diameter round, cone shaped wedges to secure each tendon, represented as round, cone shaped wedges **57A** and **57B**. In this configuration, securing three cables ($\frac{1}{2}$ " cables) at a minimum spacing would be 3.125 inches $\pm$. Given the potential minimum center to center cable spacing of 0.70" $\pm$ between individual tendon/cables in a bundled condition, the 1" diameter of the wedges prevents securing individual tendon/cables that are immediately adjacent to each other. This physical limitation means that the minimum spacing between wedge cavities in the device cannot be less than 1.40" center to center. To address this shortcoming, the '998 Patent device spaces the wedge cavities far enough apart to allow every other tendon/cable to be secured and provides enough space in between for the next immediately adjacent tendon/cable. This necessitates stacking of the devices on top of each other for securing the skipped cables.

[0070] In situations where a plurality of tendons **12** require the use of multiple adjustable lock off units **10**, the adjustable lock off units **10** may be "stacked", one of top of another/one in front of another, see FIG. 23 or FIG. 24. FIG. 23 illustrates two lock-off blocks, **10A** and **10B**, one of top of another/one in front of another, thereby providing for multiple tendons **12** to be anchored (locked-off) or non-anchored, via wedges **56**, to either lock-off block, **10A** and **10B**. FIG. 24 is a cross sectional representation of the two lock-off blocks, **10A** and **10B**, arranged one of top of another/one in front of another. This view illustrates tendon **12A** being anchored to lock-off block **10A**, but not to lock-off block **10B**. Tendon **12D** is shown anchored to lock-off block **10B**, but not to lock-off block **10A**. Tendons **12B**, and **12C** are anchored to both lock-off blocks **10A** and **10B**. As described previously, the center recess **26** allows for the multiple adjustable lock off units **10A/10B** to be stacked) without interference of the adjustable post-tensioned tendon engaging device **18** (wedges **56**) protruding out.

[0071] The load-bearing area of the lock-off block is in direct relation with the number of tendons to be temporarily anchored. Based on the total number of tendons, a certain amount of bearing area is provided per American Concrete Institute, Building Code Requirements for Structural Concrete (ACI 318-05) SO that the existing concrete surface is not overstressed due to compression force exerted onto it. The compression force that is produced by the tensile force from temporarily anchored tendons is concentrated at the loadbearing area (rear face) of the lock-off block. This force is then gradually distributed to the concrete at some distance from the lock-off block and some of this force includes some component that is transverse to the axis of the tendons. If the load-bearing area of the lock-off block is too small, then the transverse forces may cause internal tension in the concrete that is greater than the tensile strength of the concrete. Another reason for providing an adequate bearing surface is insufficient reinforcement in concrete where the lock-off openings are chipped for the installation of lock-off blocks. The thickness of the lock-off block is based on the number of tendons to be anchored, and the strength of the material used for fabrication. In order to reduce the cost of production, the lock-off block is shaped and dimensioned so that a readily available strength material can be used for fabrication. For multiple numbers of tendons that are in excess of five tendons, the thickness needs to be increased more than the thickness indicated in this application.

[0072] It is to be understood that while a certain form of the invention is illustrated, it is not to be limited to the specific form or arrangement herein described and shown. It will be apparent to those skilled in the art that various changes may be made without departing from the scope of the invention and the invention is not to be considered limited to what is shown and described in the specification and any drawings/figures included herein.

[0073] One skilled in the art will readily appreciate that the present invention is well adapted to carry out the objectives and obtain the ends and advantages mentioned, as well as those inherent

therein. The embodiments, methods, procedures and techniques described herein are presently representative of the preferred embodiments, are intended to be exemplary, and are not intended as limitations on the scope. Changes therein and other uses will occur to those skilled in the art which are encompassed within the spirit of the invention and are defined by the scope of the appended claims. Although the invention has been described in connection with specific preferred embodiments, it should be understood that the invention as claimed should not be unduly limited to such specific embodiments. Indeed, various modifications of the described modes for carrying out the invention which are obvious to those skilled in the art are intended to be within the scope of the following claims.

Claims

1. A cable lock off block for repairing a plurality of post-tensioned tendons, comprising: an anchor body having a first wall or surface, an opposing second wall or surface, a closed end, an open end, and at least one tapered inner wall or surface; an elongated slot extending a distance within said anchor body, said elongated slot having a width to accommodate the plurality of post-tensioned tendons under full tension; one or more adjustable post-tensioned cable engaging devices constructed and arranged to engage with the individual post-tensioned tendons, said or one more adjustable post-tensioned cable engaging device having a first flat side, and a second, opposing tapered side, wherein alignment with an adjacent one or more adjustable post-tensioned cable engaging device minimizes spacing therebetween.
2. The cable lock off block for repairing a plurality of post-tensioned tendons according to claim 1, wherein said one or more adjustable post-tensioned cable engaging device is a wedge body.
3. The cable lock off block for repairing a plurality of post-tensioned tendons according to claim 1, wherein said wedge body includes a tendon engaging end sized and shaped to correspond with the curvature or diameter of said post-tensioned tendons.
4. The cable lock off block for repairing a plurality of post-tensioned tendons according to claim 3 wherein said wedge body includes a center line, when placed with an adjacent wedge body, said distance between said center lines is 0.5 inches.
5. The cable lock off block for repairing a plurality of post-tensioned tendons according to claim 1, wherein said one or more adjustable post-tensioned cable engaging device is a first wedge body and a second wedge body.
6. The cable lock off block for repairing a plurality of post-tensioned tendons according to claim 5, wherein a distance between a center line of the first wedge body and a center line of the second wedge body is 0.50 inches.
7. The cable lock off block for repairing a plurality of post-tensioned tendons according to claim 2, wherein said wedge body is adjustably and movably positionable within said elongated slot.
8. The cable lock off block for repairing a plurality of post-tensioned tendons according to claim 2, further including a recessed area.
9. The cable lock off block for repairing a plurality of post-tensioned tendons according to claim 8, wherein said wedge body is disposed at least partially in the recessed area.
10. The cable lock off block for repairing a plurality of post-tensioned tendons according to claim 2, wherein a taper angle of the tapered side of said wedge body corresponds to a taper angle of the tapered interior surface.
11. The cable lock off block for repairing a plurality of post-tensioned tendons according to claim 1, further including a removable object secured within said slot.
12. The cable lock off block for repairing a plurality of post-tensioned tendons according to claim 1, wherein said removable object secured within said slot is a insert plate secured within said channel at or near said open end of said anchor body.
13. The cable lock off block for repairing a plurality of post-tensioned tendons according to claim

1, wherein said recess is centered transversally about a center of said anchor body and offset longitudinally to accommodate a bolt hole and a bolt placed perpendicular to said slot.
